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Ambient wind direction impacts on the performance of a CO_2 direct air capture system determined using large eddy simulations

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ABSTRACT

Direct air capture (DAC) has emerged as a promising technology for the direct removal of carbon dioxide (CO_2) from the atmosphere, offering a pathway to mitigate global warming. In this study, we employ the large eddy simulation (LES) approach to predict the combined effects of atmospheric boundary-layer wind speed and wind direction on the performance of a representative DAC system. We use LES coupled with our in-house boundary condition (atmosDacBoundary) code to calculate the total mass of CO_2 captured by the DAC system. The results demonstrate that with the introduction of wind, the DAC inflow velocity begins to fluctuate, resulting in time-varying amounts of CO_2 captured by the DAC unit. It is found that wind speed and direction have a significant impact on the annual amount of CO_2 captured. For a given wind angle, the annual CO_2 capture amount reduces with an increase in wind speed. For all investigated wind speeds, the annual CO_2 capture amount generally reduces with increasing wind angle relative to the DAC unit inlet. It is revealed that in the presence of wind, the maximum CO_2 capture is obtained when the DAC inlet is aligned with the wind direction (wind angle of 0°), while the minimum CO_2 capture amount occurs when the approaching wind angle is nearly perpendicular to the DAC unit inlet. Understanding this process is essential for optimising the location and orientation of DAC facilities, required for optimising the performance of DAC systems for large-scale deployment.

Nomenclature

Symbol	Definition	Unit
$a,b,\ c$	Model constants related to the relationship between capture rate and DAC inflow velocity	-
A_a	DAC inlet cross-sectional area	m^2
$(air)_{mw}$	Air molecular weights	g/mol
$(CO_2)_{ambient}$	Ambient CO_2 concentration	ppm
$(CO_2)_{cp}$	CO_2 capture amount	kg/s
$\overline{(CO_2)_{cp}}$	mean (time-averaged) CO_2 capture amount	kg/s
$(CO_2)_{in}$	DAC inlet CO_2 concentrations	ppm
$(CO_2)_{in_sf}$	Local scaled CO_2 concentration at each DAC inlet cell surface	-
$(CO_2)_{mw}$	CO_2 molecular weights	g/mol
$(CO_2)_{out}$	DAC outlet CO_2 concentrations	ppm
$(CO_2)_T$	Total mass of captured CO_2 during time t_{cc}	kg
D_{ms}	CO_2 molecular diffusivity	m^2/s
$D_{t,ms}$	Turbulent diffusivity	m^2/s
g	Gravitational acceleration	m/s^2
h	Enthalpy	J/kg
p	Pressure	Pa
Q	Q criterion	$1/s^2$
R	Gas specific constant	J/(kg. K)
s	Scaled CO_2 concentration	-
$\mathbf{S}$	Flow rate of strain tensor	$1/s$
S_p	Porous media pressure drop	Pa/m
t	Time	s

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(*continued*)

t_0	Time at which $(CO_2)_{cp}$ reaches statistically stationary conditions	s
t_{cc}	Time period corresponding to overall CO_2 mass removal	s
T	Temperature	K
u_i	Flow velocity in the *i*-direction	m/s
U_{in}	DAC inlet air flow velocity	m/s
U_{in_sf}	DAC inflow velocity at each DAC inlet cell surface	m/s
U_{ref}	Reference wind speed	m/s
U_x	Flow velocity component aligned with DAC lengthwise direction	m/s
Z_{ref}	Reference height	m
α	Surface roughness constant	-
α_{eff}	Effective thermal diffusivity	m^2/s
Δt	Sampling interval	s
η	Capture rate	-
θ	Angle between the wind direction and the DAC lengthwise direction	degree
ρ	Flow density	kg/m^3
τ_{ij}	Resolved viscous stress tensor	$kg/(m.s^2)$
τ_{ij}^{sgs}	Sub-filter stresses	$kg/(m.s^2)$
Φ_{sf}	DAC inlet cell surface contribution to DAC outlet CO_2 concentration	-
$\boldsymbol{\Omega}$	Rate of rotation tensor	$1/s$
$\overline{()}$	Reynolds filters	
$\widetilde{()}$	Favre filter	
Abbreviations		
ABL	Atmospheric boundary layer	
CDR	Carbon dioxide removal	
CO_2	Carbon dioxide	
DAC	Direct air capture	
LES	Large eddy simulation	
VOF	Volume-of-fluid	

1. Introduction

The ongoing surge in atmospheric carbon dioxide (CO_2) levels as the major greenhouse gas associated with human activities, is a key driver of global warming. Scientific agreements consider rapid decarbonisation as a solution to limit temperature increases below 2 °C by 2100 (Intergovernmental Panel on Climate Change, 2023). Model predictions demonstrate that emission reduction strategies alone are not enough to reach net-zero by 2050. Achieving this target requires advancing carbon dioxide removal (CDR) technologies for large-scale deployment with an estimated annual CO_2 removal capacity of nearly 3.1 Gt (IEA, 2022). As a viable carbon removal technology, Direct air capture (DAC) systems employ physical or chemical separation techniques to extract CO_2 from ambient air at large scale. Effective large-scale DAC deployment requires optimal DAC designed to operate efficiently under diverse atmospheric conditions while ensuring minimal negative impacts on the local environment. DAC systems are based on either liquid absorbents or solid sorbent for CO_2 capture. In the liquid-based systems, a typical approach is the use of aqueous potassium hydroxide solutions as suggested by Carbon Engineering (Holmes and Keith, 2012; Keith et al., 2018) or using amine-based solutions in commercial cooling towers, as proposed by Abdellah et al. (2024) and Kiani et al. (2021).

The use of industrial cooling tower geometry as the gas-liquid contactor for DAC applications has gained interest because of the availability of inexpensive components, such as packing materials, fans, and liquid distribution equipment, making them a cost-effective option (Abdellah et al., 2024; Holmes and Keith, 2012; Keith et al., 2018; Kiani et al., 2020, 2021). In cooling tower-based DAC technologies, adopted as the DAC geometry in the present study, the absorption liquid is distributed across the packing material, capturing CO_2 from incoming air entrained by a fan. The exhaust fan then releases CO_2-depleted air into the surrounding air, while the CO_2-rich solution is collected in a basin for the subsequent regeneration process.

Computational fluid dynamics (CFD) has become a key tool to simulate carbon capture processes in the packing materials as well as DAC systems. A multiphase Eulerian CFD approach was used to model CO_2 capture processes to obtain the absorber performance associated with an amine-based CO_2-absorbent system (Kim et al., 2016; Niegodajew and Asendrych, 2016; Pham et al., 2015). Elsewhere, the volume-of-fluid (VOF) approach associated with multiphase CFD modelling was employed to model gas-liquid mass transfer within structured packing elements (Sebastia-Saez et al., 2013, 2014, 2015). Other studies also considered CFD simulations to predict flow behaviour inside structured packing used in CO_2 absorption columns (Basha et al., 2020; Haroun et al., 2014, 2012).

Studies that employ CFD simulations for DAC performance assessment are, however, relatively limited. For instance, one study examined the impacts of adsorption bed geometry on both CO_2 capture performance and the energy demand of an indoor DAC system (Chen et al., 2023). Elsewhere, CFD was used to numerically assess gyroid-type triply periodic minimal surface (TPMS) contactors for direct air capture applications (Kincaid et al., 2025). Another study employed CFD simulations to evaluate wind energy usage to drive airflow in DAC systems, concentrating on the effects of wind-catcher geometry as well as wind speed on the overall DAC performance (Akinjide et al., 2023). In the same vein, a CFD study proposed integration of a DAC system with floating wind turbines (Foxall et al., 2024), demonstrating that the use of wind-driven flow can lower fan energy demand in that system. More recently, Pedrozo et al. (2025) conducted a CFD-based study for the DAC adsorption-desorption cycle optimisation, highlighting the effects of contributing parameters such as reactor design, bed porosity, and operating conditions on the optimisation process.

In our recent study, we employed a high-fidelity turbulence-resolved large eddy simulation (LES) approach to resolve the interaction between ambient wind with a CO_2-depleted plume exiting a DAC unit (Eftekharian et al., 2025a). This led to the development of practical statistical relationships for the CO_2 concentration as a function of distance downstream of a DAC unit and the DAC exit-wind velocity ratio. In another recent study (Eftekharian et al., 2025b), we have developed a new boundary condition model that enables a CFD-based numerical model to predict a DAC system performance behaviour under realistic environmental conditions, such as the impact of ambient wind. This new model couples external flow dynamics such as wind and turbulence with the internal performance of the DAC systems based on the macro-scale relationships between capture rate and DAC inflow velocity obtained from laboratory experiment. This new numerical model was developed and implemented in OpenFOAM CFD platform and is capable of dynamically calculating the time-varying fraction of CO_2 removed from the air (capture rate) and the total mass of CO_2 captured by the system per unit time (capture amount) under different ambient wind conditions (Eftekharian et al., 2025b).

Nonetheless, the previous study (Eftekharian et al., 2025a) assessed the impacts of ambient wind speed on a DAC system performance for a single wind direction only. The influence of wind direction on DAC performance behaviour, which is of practical significance in real environmental conditions, remains unexplored. In particular, the mechanisms through which wind direction causes changes in the DAC inlet flow characteristics, interaction between wake recirculation and DAC system, and DAC performance remain unclear. As such, to understand the impacts of the ambient wind direction on DAC performance and to identify the corresponding underlying physical mechanisms, a systematic analysis of these coupled effects is required. Unlike the previous single wind direction studies (Eftekharian et al., 2025a, b), the current study investigates multiple wind speeds and wind directions to identify how wind-direction-dependent wake interaction with the DAC inlets affects overall CO_2 capture performance of a DAC system under realistic environmental conditions. As such, we use our in-house boundary condition (atmosDacBoundary) solver associated with LES simulations to study the impacts of wind direction on DAC unit performance. Different combinations of wind directions and speeds are simulated, and

the corresponding impacts on the CO_2 capture amount and capture rate are quantitatively calculated.

2. Model description

2.1. Governing equations

The thermo-fluid governing equations were numerically solved using our in-house developed solver on the OpenFOAM platform (Eftekharian et al., 2025a, b). This solver is based on sprayFoam solver of OpenFOAM, which is an LES-based Eulerian-Lagrangian solver. For the present study, the Lagrangian particle tracking, reacting flow and combustion capabilities of this solver were disabled, leaving only the non-reacting, continuous gas-phase component active. Continuity, momentum, enthalpy and state equations were solved using our density-variable, low-Mach number solver:

$$\frac{\partial \overline{\rho}}{\partial t} + \frac{\partial \overline{\rho}\widetilde{u}_j}{\partial x_j} = 0 \tag{1}$$

$$\frac{\partial \overline{\rho}\widetilde{u}_i}{\partial t} + \frac{\partial \overline{\rho}\widetilde{u}_i\widetilde{u}_j}{\partial x_j} = -\frac{\partial \overline{p}}{\partial x_i} + \frac{\partial \tau_{ij}}{\partial x_j} + \frac{\partial \tau_{ij}^{sgs}}{\partial x_j} + \overline{\rho} g_i + \widetilde{S}_p \tag{2}$$

$$\frac{\partial \overline{\rho}\widetilde{h}}{\partial t} + \frac{\partial \overline{\rho}\widetilde{u}_j\widetilde{h}}{\partial x_j} = \frac{D\overline{p}}{Dt} + \frac{\partial}{\partial x_j}\left(\overline{\rho}\alpha_{\text{eff}}\frac{\partial \widetilde{h}}{\partial x_j}\right) \tag{3}$$

$$\overline{p} = \overline{\rho} R \widetilde{T} \tag{4}$$

a repeated index indicates Einstein's summation convention, where u_i is the flow velocity in the i-direction. g, p, and ρ are the gravitational acceleration, flow pressure, and density, respectively. τ_{ij} is the resolved viscous stress tensor, and $\tau_{ij}^{sgs} = \overline{\rho}\widetilde{u}_i\widetilde{u}_j - \overline{\rho}\widetilde{u}_i\widetilde{u}_j$ denotes the sub-filter stresses, which are treated using a kEq model (Yoshizawa, 1986). h is the enthalpy, R is the gas specific constant, T is the temperature, and α_{eff} is the effective thermal diffusivity, which incorporates both molecular and turbulent contributions (Eftekharian et al., 2025a). S_p is the (negative) porous media pressure drop source term associated with the Darcy Fochheimer model, which was described in detail in our previous work (Eftekharian et al., 2025a). Superscript "$\sim$" and "$-$" denote Favre and Reynolds filters, respectively.

The CO_2 dispersion was modelled using a scalar transport equation as:

$$\frac{\partial \overline{\rho}\widetilde{s}}{\partial t} + \frac{\partial \overline{\rho}\widetilde{u}_j\widetilde{s}}{\partial x_j} = \frac{\partial}{\partial x_j}\left(\overline{\rho}\left(D_m + D_{t,ms}\right)\frac{\partial \widetilde{s}}{\partial x_j}\right) \tag{5}$$

where s is the CO_2 concentration, D_{ms} and $D_{t,ms}$ are the CO_2 molecular diffusivity and turbulent diffusivity, respectively.

2.2. Geometry and boundary conditions

The computational domain includes the DAC unit within its surrounding environment as shown in Fig. 1. The DAC system geometry is based on the existing cooling tower designs (Eftekharian et al., 2025a), adjusted with extended porous fill-packing sections at each air inlet. This

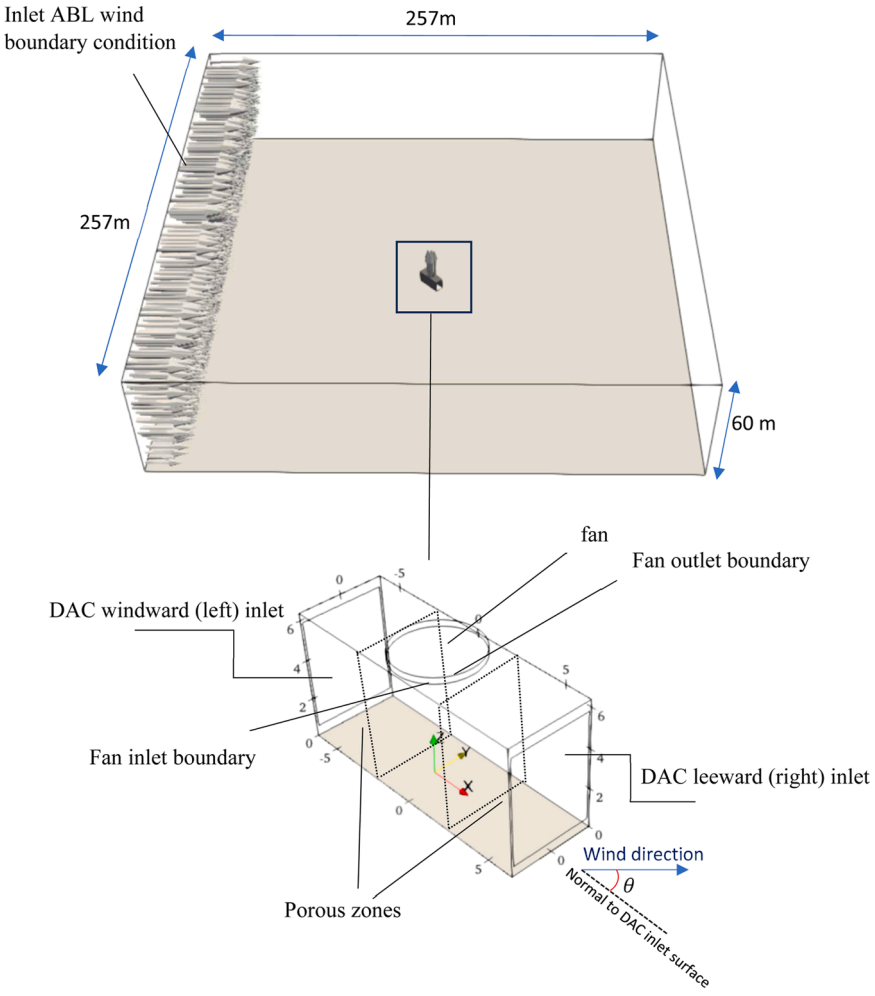


Fig. 1. Computational domain (top diagram) and magnified view of the DAC unit (bottom diagram). All dimensions are in meters.

system includes a central fan at the top and two porous fill-pack regions, each 4.5 m long, connected to the inlet sides.

The fan is modelled by imposing a constant vertical velocity on both sides of the fan region. The fan vertical velocity was adjusted to 3.347 m/s, drawing a flow rate of 43.55 m^3/s into the DAC unit through the two inlets and discharging the air through the fan outlet into the environment. This boundary condition induces a DAC inlet velocity of approximately 1 m/s under still (no-wind) ambient conditions. The selection of these parameters is based on laboratory-scale design assumptions of a fixed capture rate (50 %) at a fixed inlet air velocity of 1m/s, which was shown to be a cost-effective parameter choice (Eftekharian et al., 2025b; Kiani et al., 2020). Likewise, in the present study, selecting a fan velocity of 3.347 m/s leads to a DAC inlet superficial velocity of 1 m/s in the design condition, which signifies a balanced operating condition: lower velocities would require the treatment of significantly larger air volumes, whereas higher velocities could result in a lower capture rate due to less interaction between the solvent solution particles and the DAC inflow (Eftekharian et al., 2025b; Kiani et al., 2020). Overall, selecting the vertical velocity of 3.347 m/s as the fan boundary condition induces a superficial bed velocity of 1 m/s in the design condition, which results in achieving about 50 % capture rate, which is the balanced-design point (Eftekharian et al., 2025b; Kiani et al., 2020).

No-slip wall boundary conditions were used for the ground and DAC walls, whereas the slip boundary condition was applied at the domain sides and top boundaries. The ambient wind was modelled using an atmospheric boundary layer (ABL) wind profile, represented by a power-law variation with height, as employed in our earlier studies (Eftekharian et al., 2020a, 2019a, 2020b; Eftekharian et al., 2019b, 2025a, b; Eftekharian et al., 2021).

$$U_z = U_{\mathrm{ref}}\left(\frac{Z}{Z_{\mathrm{ref}}}\right)^{\alpha} \tag{6}$$

where U_{ref} denotes a reference wind speed at a reference height of $Z_{\mathrm{ref}} =$ 6.35 m. We consider different reference wind speeds of $U_{\mathrm{ref}} = 1$ m/s, 2 m/s, 3 m/s, 5 m/s, 6 m/s, 7 m/s and 9 m/s, covering light to strong winds. The exponent α is a constant empirical parameter determined using both surface roughness and atmospheric stability characteristics. In this study, the exponent value of $\alpha = 0.16$ is used, which corresponds to the atmospheric neutral stability condition. To identify the effects of wind direction on the DAC CO_2 capture characteristics, simulation scenarios with different wind angles were performed for all the above-mentioned wind speeds. Here, the wind angle θ is defined as the angle between the wind direction and the lengthwise direction of the DAC unit, as shown in Fig. 1. So, an angle of 0° means the DAC inlet is facing the wind. Five different wind angles ($\theta = 0$, 30, 45, 60, and 90 degrees) were considered in the current simulations. The global coordinate system is attached to the DAC unit in all wind angle simulation scenarios, where the x-direction is always normal to the DAC inlets. The computational domain illustrated in Fig. 1 corresponds to the wind angle of θ=45° which includes nearly 17 million cells. For other wind directions, the domain dimensions were adjusted to minimise boundary influence while optimising computational cost. In all cases, the grid resolution around the DAC unit was kept similar, though the total cell numbers altered due to different domain extents. The chosen grid resolution used in the current simulations was verified as grid-independent in our previous works (Eftekharian et al., 2025b). This selected grid resolution satisfies the Pope criterion (Pope, 2001) for an LES simulation, confirming its appropriateness for resolving >80 % of the total turbulent kinetic energy (Eftekharian et al., 2025b).

It was shown in our previous study that CO_2 capture rate and capture amount depend on the DAC unit inlet flow velocity, and as such, the CO_2 concentration at the DAC unit outlet is a function of DAC unit inlet velocity (Eftekharian et al., 2025b).

The capture rate (η) and capture amount $(CO_2)_{cp}$ are defined as the fraction of CO_2 removed from the air, and the total amount of CO_2 captured by the system (kg/s), respectively (Eftekharian et al., 2025b):

$$\eta = ((CO_2)_{\mathrm{in}} - (CO_2)_{\mathrm{out}})/(CO_2)_{\mathrm{in}} = 1 - (CO_2)_{\mathrm{out}}/(CO_2)_{\mathrm{in}} \tag{7}$$

$$(CO_2)_{\mathrm{cp}} = \rho U_{\mathrm{in}} A_a((CO_2)_{\mathrm{in}} - (CO_2)_{\mathrm{out}}) \times 10^{-6} \times \frac{(CO_2)_{\mathrm{mw}}}{(\mathrm{air})_{\mathrm{mw}}} \tag{8}$$

Combining Eq. (7) and 8 leads to:

$$(CO_2)_{\mathrm{cp}} = \rho U_{\mathrm{in}} A_a (CO_2)_{\mathrm{in}} \eta \times 10^{-6} \times \frac{(CO_2)_{\mathrm{mw}}}{(\mathrm{air})_{\mathrm{mw}}} \tag{9}$$

where $(CO_2)_{\mathrm{in}}$ and $(CO_2)_{\mathrm{out}}$ denote CO_2 concentrations (ppm) at the DAC inlet and outlet, respectively. ρ is the air density, U_{in} is the inlet air velocity (m/s), and A_a is the inlet cross-sectional area (m^2) of the DAC unit. $(\mathrm{air})_{\mathrm{mw}}$ and $(CO_2)_{\mathrm{mw}}$ are the molecular weights of air and CO_2, respectively.

The overall CO_2 mass removal over a time period of t_{cc} (s) can be obtained by using the mean (time-averaged) CO_2 capture amount $\overline{(CO_2)_{cp}}$:

$$(CO_2)_T = \overline{(CO_2)_{\mathrm{cp}}} \times t_{cc} \tag{10}$$

where $(CO_2)_T$ denotes the total mass (kg) of captured CO_2 during time t_{cc} (s). The mean (time-averaged) CO_2 capture amount, $\overline{(CO_2)_{cp}}$, is obtained through temporal integration of the instantaneous CO_2 capture amount (Eq. (9)) over the sampling interval (Δt):

$$\begin{aligned}\overline{(CO_2)_{cp}} &= \frac{1}{\Delta t}\int_{t_0}^{t_0+\Delta t} (CO_2)_{\mathrm{cp}}(t)dt \\ &= \frac{1}{\Delta t}\int_{t_0}^{t_0+\Delta t} \left(\rho U_{\mathrm{in}}(t) A_a (CO_2)_{\mathrm{in}}(t)\eta(t) \times 10^{-6} \times \frac{(CO_2)_{\mathrm{mw}}}{(\mathrm{air})_{\mathrm{mw}}}\right)dt\end{aligned} \tag{11}$$

where, $\overline{(CO_2)_{cp}}$ denotes the mean CO_2 capture amount time-averaged over Δt period, and t_0 is the time at which $(CO_2)_{\mathrm{cp}}(t)$ reaches statistically stationary conditions. Eqs. (10)–11 reveal that the total CO_2 capture amount is obtained based on temporal integration of the instantaneous CO_2 capture amount over the time averaging interval rather than from products of independently time-averaged variables. As such, the effects of CO_2 capture characteristics and the flow fluctuations are inherently included in the mean values calculation.

CO_2 distribution was modelled using Eq. (5) together with an in-house developed time-dependent boundary for the DAC exit CO_2 concentration. The CO_2 transport equation (Eq. (5)) was solved at each time step and coupled with the instantaneous velocity field obtained from the flow solution. A bounded second-order scheme was employed for spatial discretisation of the convective term to keep numerical stability while maintaining solution accuracy. A second-order linear interpolation approach was employed to discretise the gradient term, while a second-order corrected scheme was used for the diffusion term to account for any mesh non-orthogonality. The in-house developed boundary condition (atmosDacBoundary) (Eftekharian et al., 2025b) was used as the boundary condition of the CO_2 transport equation for the fan outlet. This boundary condition instantaneously adjusts the DAC exit CO_2 concentration based on the capture rate and capture amount at each time step (Eftekharian et al., 2025b). From Eq. (7), $(CO_2)_{\mathrm{out}}$ can be written as a function of the CO_2 capture rate (η):

$$(CO_2)_{\mathrm{out}} = (CO_2)_{\mathrm{in}}(1-\eta), \tag{12}$$

It was shown in our previous study that η is a quadratic function of the DAC inlet flow velocity (Eftekharian et al., 2025b):

$$\eta = aU_{\mathrm{in}}^2 + bU_{\mathrm{in}} + c \tag{13}$$

in which, U_{in} denotes the DAC unit inlet flow velocity (m/s), which is equivalent to the velocity component normal to the unit inlet surface and *a, b*, and *c* are constants with values of $a = 0.0672$, $b = -0.3269$, and $c = 0.7588$, respectively. Eq. (13) indicates that the CO_2 capture rate is a rank 2 polynomial function of DAC inlet velocity. It was also shown that this relationship between capture rate and inlet velocity (Eq. (13)) remains valid for a small variation (within 10 %) of DAC inlet CO_2 concentration (Eftekharian et al., 2025b).

As such $(CO_2)_{out}$ can be written as a function of CO_2 concentration at the unit inlet and unit inlet velocity by substituting Eq. (13) into Eq. (12):

$$(CO_2)_{out} = (CO_2)_{in}\left(1 - \left(aU_{in}^2 + bU_{in} + c\right)\right) \tag{14}$$

This boundary condition fully couples CO_2 concentration at the DAC outlet with the DAC inflow velocity and CO_2 concentration at the DAC inlet, as detailed in Eq. (14). To demonstrate numerical implementation of this relationship, we define the variable Φ_{sf}:

$$\Phi_{sf} = (CO_2)_{in_sf}\left(1 - \left(aU_{in_sf}^2 + bU_{in_sf} + c\right)\right) \tag{15}$$

where Φ_{sf} is calculated locally at each DAC inlet surface cell. U_{in_sf} and $(CO_2)_{in_sf}$ indicate the local DAC inflow velocity and CO_2 concentration at each DAC inlet cell surface, respectively. The $(CO_2)_{out}$ in Eq. (14) is then calculated using a surface-area-weighted averaging of Φ_{sf} over the surface cells across the DAC inlet regions.

It should be noted that any CO_2-depleted air recirculation associated with the DAC exit back into the inlet region is naturally captured through the instantaneous DAC inflow velocity and CO_2 concentration fields obtained from the flow solution. Consequently, CO_2-depleted plume recirculation effects are inherently considered through the DAC instantaneous inflow and CO_2 capture characteristics used in the $(CO_2)_{out}$ calculation.

We developed this boundary condition in OpenFOAM, which has the advantage of resolving the two-way interaction between the DAC unit CO_2 capture characteristics as well as the CO_2-depleted exit plumes in a real complex atmospheric flow environment (Eftekharian et al., 2025b). Therefore, this boundary condition can resolve the potential impacts of vortical structures, formed around the DAC unit inlets due to the DAC-wind interaction, on the CO_2 capture characteristics. The time history of CO_2 capture characteristics (capture rate and capture amount) for each simulated flow configuration was also instantaneously calculated and stored by adding post-processing code to OpenFOAM.

3. Validation

Our in-house LES solver was validated against different sets of experimental data, direct numerical simulation (DNS), and analytical data associated with various jets in crossflow (i.e. wind with a vertical plume). One of the datasets used in this regard was the Chalk Point Dye Tracer Experiment (Davis, 1979; Hanna, 1978; Policastro et al., 1978, 1981; Policastro and Wastag, 1981) associated with exit plume characteristics from the cooling tower under atmospheric wind conditions. Other studies include the jet-in-crossflow results associated with direct numerical simulations by Muppidi and Mahesh (2005) and analytical formula presented by Kamotani and Greber (1972). The details of these validation studies are demonstrated in our previous work (Eftekharian et al., 2025a), where our LES results exhibited very good agreement with the above-mentioned experimental data, DNS and analytical results.

In addition to these validation studies, we benchmarked our atmosDacBoundary using our own experimental data (Eftekharian et al., 2025b) described in Section 2. We used an amine-based DAC unit to predict the corresponding carbon capture performance (Fig. 2). In this configuration, a commercial counter-current cooling tower was used as the air contactor (Kiani et al., 2022). The cooling tower includes a packing height of 0.6 m, an inlet area of 0.6 × 0.6 m^2, and a 100 L reservoir at the base. This system recirculates the liquid to the top for distribution over the packing. Air with the CO_2 concentration of nearly 420 ppm (an approximate present-day's CO_2 concentration) is drawn into the unit by a fan at the top while reacting with the absorbent solution flowing over the packing, causing a reduction in the CO_2 concentration in the exiting air.

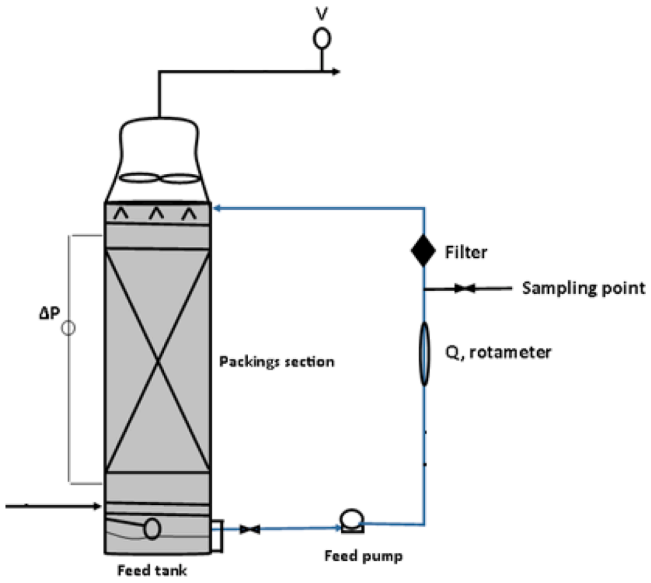


Fig. 2. Schematic of the cooling tower contactor (Eftekharian et al., 2025a).

A schematic view of the setup is shown in Fig. 2. The fan speed was adjusted such that different air velocities ranging from 0 to 2 m/s were obtained to evaluate the impacts of air flow velocity on the capture performance of the DAC unit. For consistency, in all experiments, the absorbent solution flow rate was maintained constant at 40 L/min.

The CO_2 capture is characterised in terms of CO_2 capture rate and capture amount, which were experimentally measured for different unit inlet velocities (Fig. 3). CO_2 capture characteristics are dependent on the flow velocity inside the DAC unit (Fig. 3), and this relation can be described as a rank two polynomial function (Eq. (13)). We used this relationship to inform the boundary condition for the CO_2 concentration at the unit outlet as introduced in Section 2.

During the experiments, DAC inlet CO_2 concentration varied between 380 ppm and 420 ppm, corresponding to a nearly 10 % variation relative to the ambient reference CO_2 concentration (420 ppm). The correlation demonstrated in Eq. (13) was derived and calibrated based on the experimental measurements obtained within this operating range of CO_2 concentration. The relationship between the unit inlet velocity and capture rate in Eq. (13) is therefore considered applicable and reliable for the studied 10 % variation range.

For numerical model validation purposes, a set of LES simulations was performed for the DAC geometry described in Fig. 1, but in the absence of wind to replicate still air (no-wind) conditions similar to the experiment. Different simulation scenarios with different fan velocities were performed to replicate DAC unit inlet velocities obtained from the experiment, as shown in Fig. 3. For the validation case, open boundary (pressureInletOutletVelocity) and atmospheric pressure conditions were employed for the velocity and pressure of the surrounding domain and the top boundaries. A no-slip wall boundary condition was used for the DAC unit walls and the ground.

It should be highlighted that the performed validation study was not intended to demonstrate direct geometric equivalence between the laboratory-scale counter-current DAC system and the full-scale cross-flow DAC unit considered in the numerical simulations. Instead, the validation aimed to evaluate the capability of the developed numerical

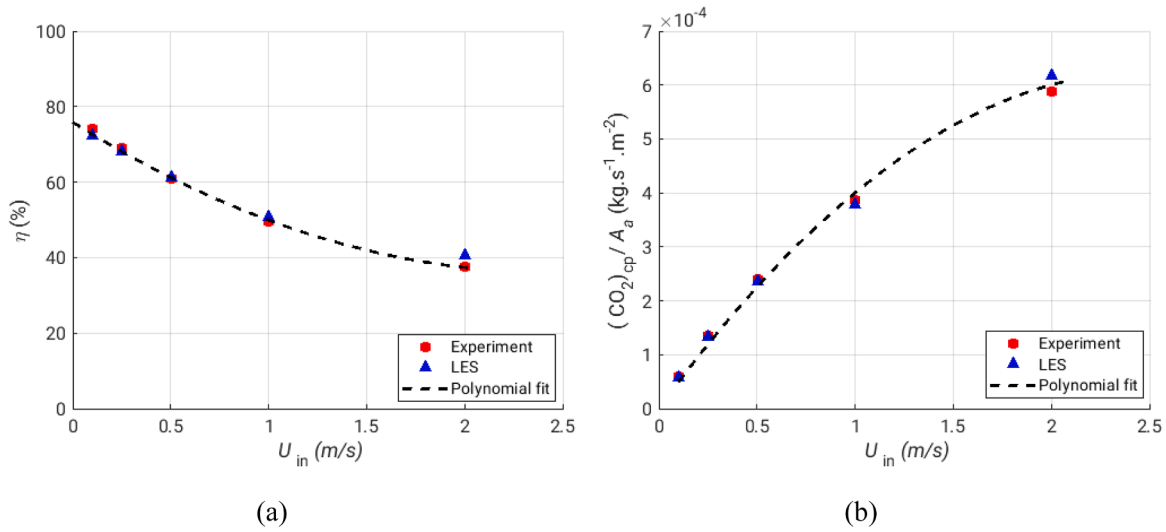


Fig. 3. Comparison of the LES and the experiment for different unit inlet velocities (U_{in}) on the basis of: (a) CO_2 capture rate (η) (%); and (b) CO_2 capture amount per inlet cross-section area ($(CO_2)_{cp}/A_a$) (Eftekharian et al., 2025b).

modelling to reproduce the CO_2 capture-airflow behaviour under comparable operating conditions. In both laboratory and full-scale DAC cases, inlet airflow velocity and the pressure-drop characteristics associated with the packing material, the two key parameters affecting the DAC CO_2 performance, were maintained consistently. Consequently, the validation provides a meaningful evaluation of the numerical model accuracy despite the geometric and configuration differences between the experimental set-up and the numerical model.

The predicted CO_2 capture rate and capture amount obtained from the LES solutions show an excellent agreement (average error <3 %) with the experimental measurements (Fig. 3) (Eftekharian et al., 2025b), confirming the accuracy of our in-house-developed CO_2 boundary condition (atmosDacBoundary) in predicting CO_2 capture characteristics. The LES prediction of pressure drop associated with the porous packing areas on each side of the DAC was also compared with the experimental data, with an average error of <5 % as reported in our previous work (Eftekharian et al., 2025a).

4. Results

4.1. Visualisation of the instantaneous three-dimensional fields

Fig. 4 visualises a 3D view of flow and CO_2 concentration around a DAC unit at different wind directions using isosurfaces of the quantity Q (left column) and CO_2 concentration (right column). The Q criterion is a vortex-identification approach that visualises vortex structures in turbulent flows (Jeong and Hussain, 1995; Zhan et al., 2019):

$$Q = \frac{1}{2}\left(\| \boldsymbol{\Omega}\|^2 - \| \mathbf{S}\|^2 \right), \tag{16}$$

where, $\mathbf{S}$ is the flow rate of the strain tensor and $\boldsymbol{\Omega}$ the rate of rotation tensor. $Q > 0$ in a region represents the area where the flow vorticity magnitude is greater than the magnitude of the rate of strain, indicating the presence of a vortex in that region (Zhan et al., 2019). Visualising surfaces of Q near zero indicates the boundary of a vortex.

Fig. 4 shows that the wind direction significantly affects the mechanisms of vortical structure formation around the DAC unit. With the change of wind direction, the vortical structure near the DAC unit inlet changes. At $\theta = 0°$, the dominant vortex structures near the DAC inlets include horseshoe vortices and wake-induced vortices. At $\theta = 90°$, streamwise edge vortices are dominant, formed due to the flow separation associated with wind striking vertical edges of the DAC wall. The presence of these vortical structures near the DAC inlets and their variation with wind direction can affect the corresponding DAC unit inlet velocity, which, according to Eqs. (7)–9, result in changes in CO_2 capture characteristics. It should be noted that in all Fig. 4's subfigure, the DAC orientation and viewing perspective are kept fixed for consistency and ease of comparison, while the ambient wind direction is schematically shown using the indicated wind-direction arrow and angle. Consequently, in most cases, the windward inlet remains visible while the leeward inlet is located on the rear side of the displayed view. The case with the wind angle of $\theta = 90°$ is a special configuration where the distinction between leeward and windward inlets becomes less physically meaningful.

In addition to the DAC unit inlet velocity, wind direction significantly impacts the distribution of the CO_2-depleted exit plume (Fig. 4). The isosurface contour of CO_2 corresponds to those regions where the CO_2 concentration is lower than that of the ambient air ($CO_2 \leq 0.95(CO_2)_{ambient}$), representing the region where the CO_2 concentration of the surrounding environment is impacted by the CO_2-depleted exit plume. At $\theta = 0°$, and lower oblique angles (e.g. $\theta = 30°$ and $\theta = 45°$), a portion of the DAC CO_2-depleted exit plume is drawn back to the unit through the leeward inlet as it mixes with the wake recirculation zone formed near that inlet. CO_2-depleted plume re-entry back into the DAC unit inlet decreases with increasing wind angle, becoming nearly zero at θ= 90°. This is because at $\theta = 90°$, the wake region responsible for the recirculation of the CO_2-depleted plume forms predominantly near the DAC wall rather than in the region near the DAC inlet, preventing the CO_2-depleted plume from re-entering the unit (Fig. 4). As such, wind direction affects CO_2 concentration at the DAC unit inlet, leading to changes in CO_2 capture characteristics (Eqs. (7)–9) with variation in wind direction.

4.2. Statistical description

To further statistically understand the impacts of wind speed and direction on CO_2 capture characteristics, the distribution of flow velocity and CO_2 concentration (scaled by the ambient CO_2 concentration) near the DAC unit inlet at different wind angles for a given wind speed is plotted in Figs. 5–8. The presence of wind deflects the CO_2-depleted plume aligned with the wind direction. For all wind angles of θ (where θ is the angle between the wind direction and the direction normal to the DAC unit inlet (see Figs. 1 and 4)), the CO_2-depleted plume is curved due to the high horizontal momentum. This mixes the CO_2-depleted plume with the wake region formed downstream of the DAC unit, due to the wind-DAC interaction. For the wind angle $\theta = 0°$, where the wind is

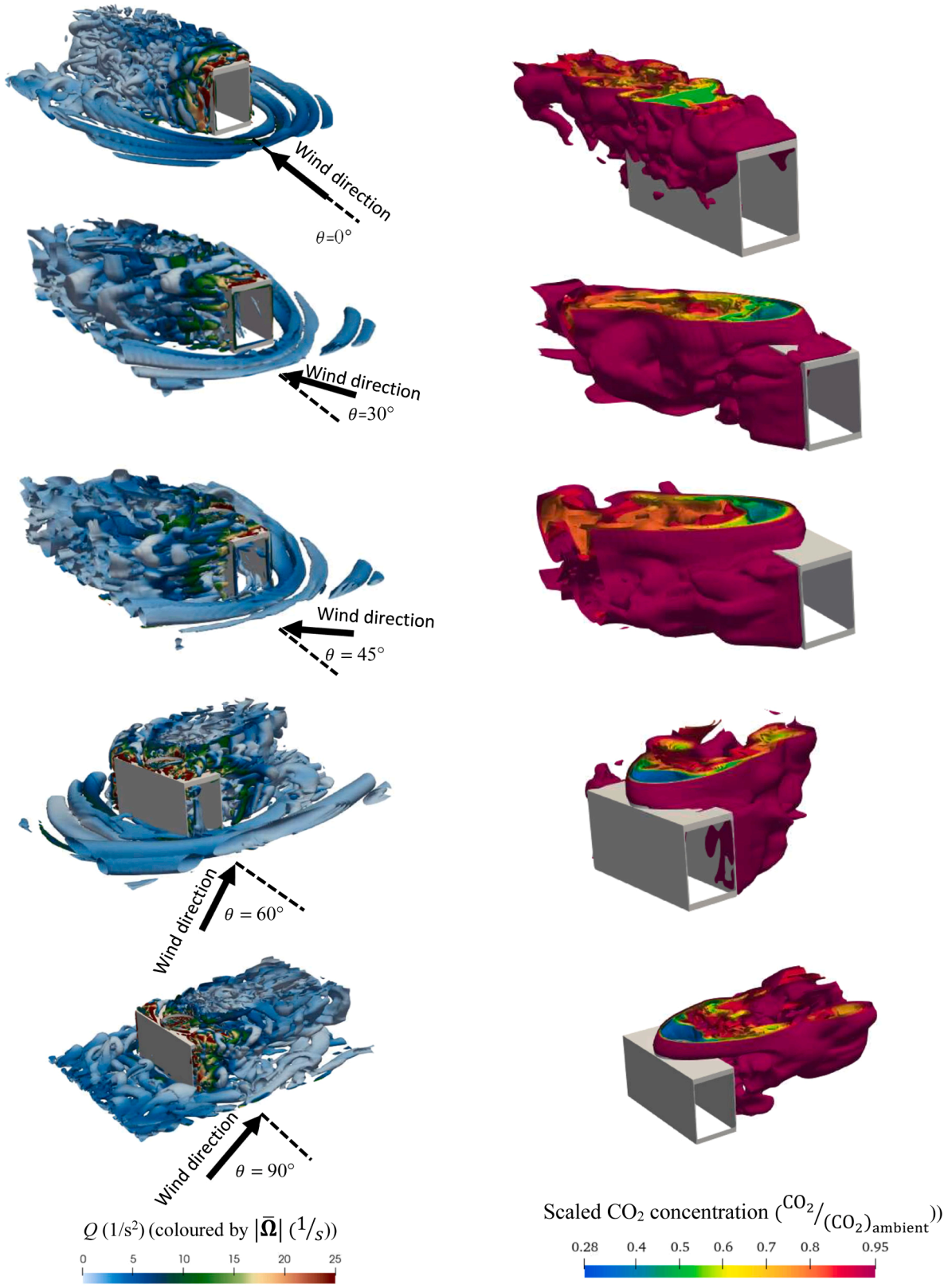


Fig. 4. Iso-surface of Q-criterion ($1/s^2$) associated with velocity fluctuations (left column ranging between 0.12 and 1.2 and coloured by mean vorticity magnitude) and isosurface of scaled CO_2 concentration (right column coloured by instantaneous scaled CO_2 concentration) for different wind angles ($\theta = 0°$, 30°, 45°, 60°, 90°). Wind reference velocity is $U_{ref} = 7$ m/s for all cases. The angle shown in the figure represents the angle between the wind direction (solid arrow) and unit inlet direction (dash line). For consistency and comparative visualisation of the flow structures, the DAC orientation and viewing perspective are kept fixed for all subfigures, while the incoming wind direction is varied according to the indicated wind-direction arrow and angle.

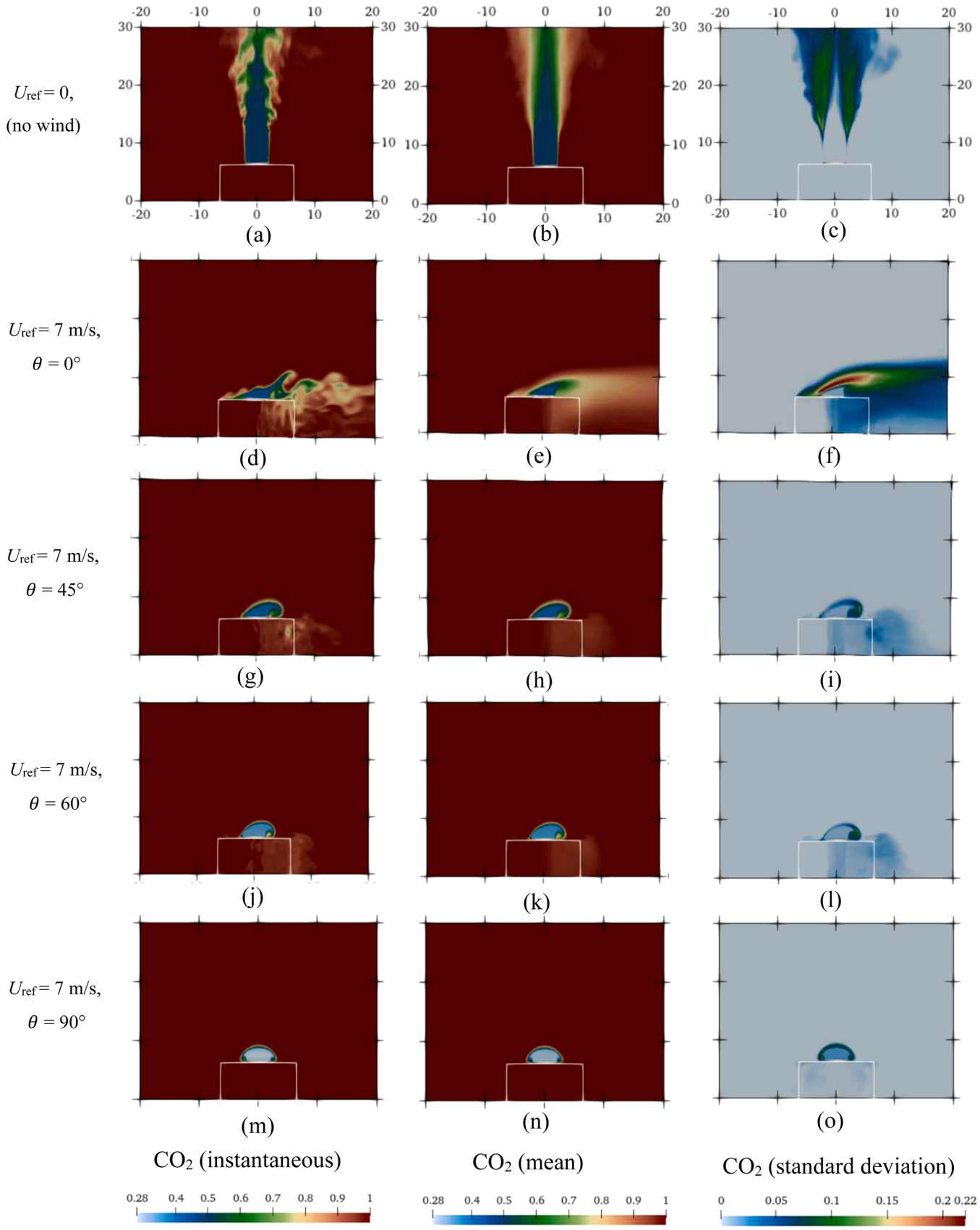


Fig. 5. Distribution of scaled CO_2 concentrations including instantaneous (at $t = 155$ s) (left column) mean (middle column), and standard deviation (right column) for different wind angles along a vertical plane passing through the DAC centreline ($y = 0$) for both the no-wind condition and wind reference velocity of $U_{ref} = 7$ m/s.

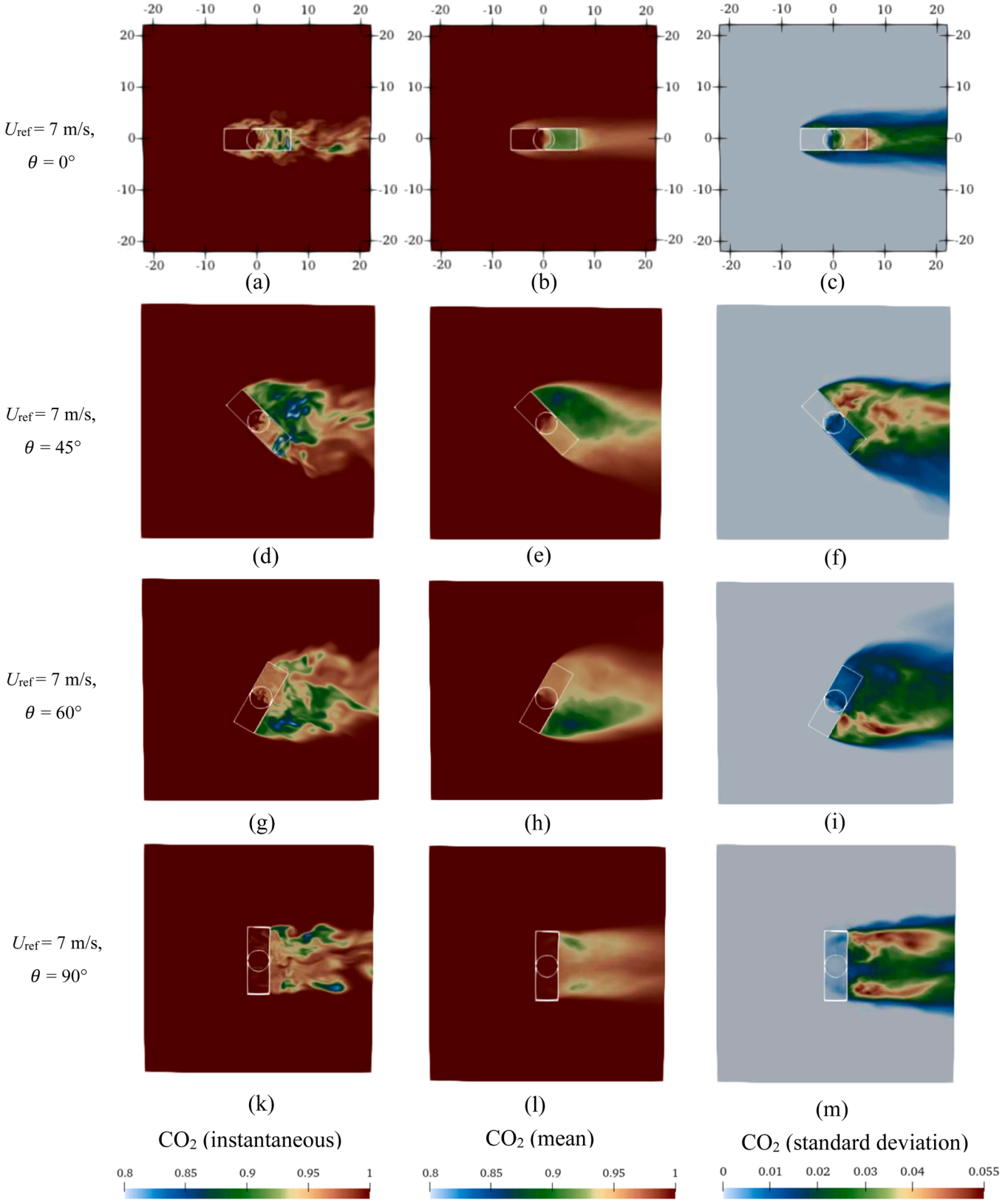


Fig. 6. Distribution of scaled CO_2 concentrations including instantaneous (at t = 155 s) (left column) mean (middle column), and standard deviation (right column) for different wind angles at a horizontal plane passing through the DAC unit middle height (z = 3.175 m) for the wind reference velocity of U_{ref} = 7 m/s.

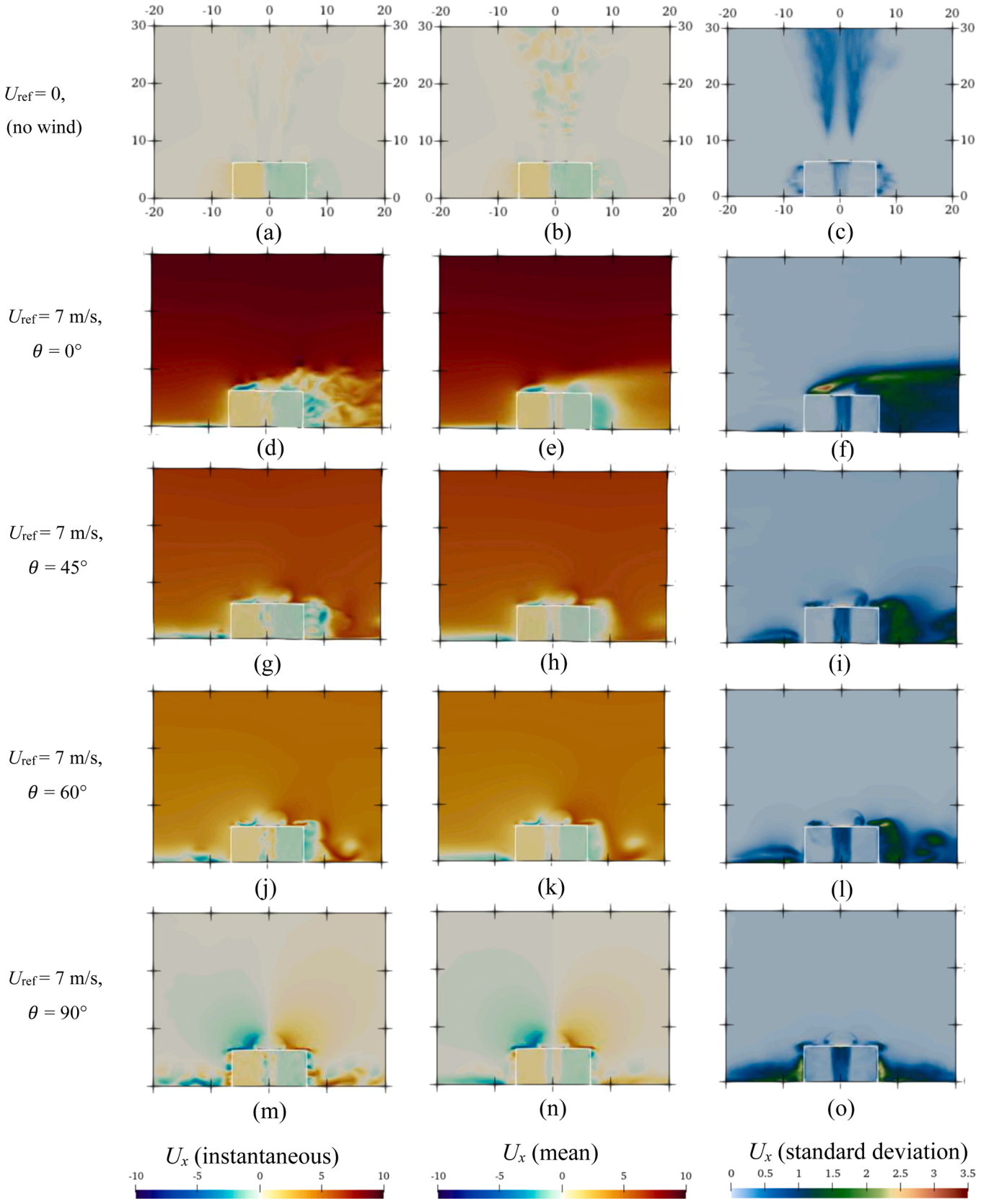


Fig. 7. Distribution of flow velocity coloured by the velocity component aligned with the DAC lengthwise direction (U_x) (m/s), including instantaneous (at $t = 155$ s) (left column) mean (middle column) and standard deviation (right column) for different wind angles along a vertical plane passing through the DAC centreline ($y = 0$) for both no-wind condition and wind reference velocity of $U_{ref} = 7$ m/s.

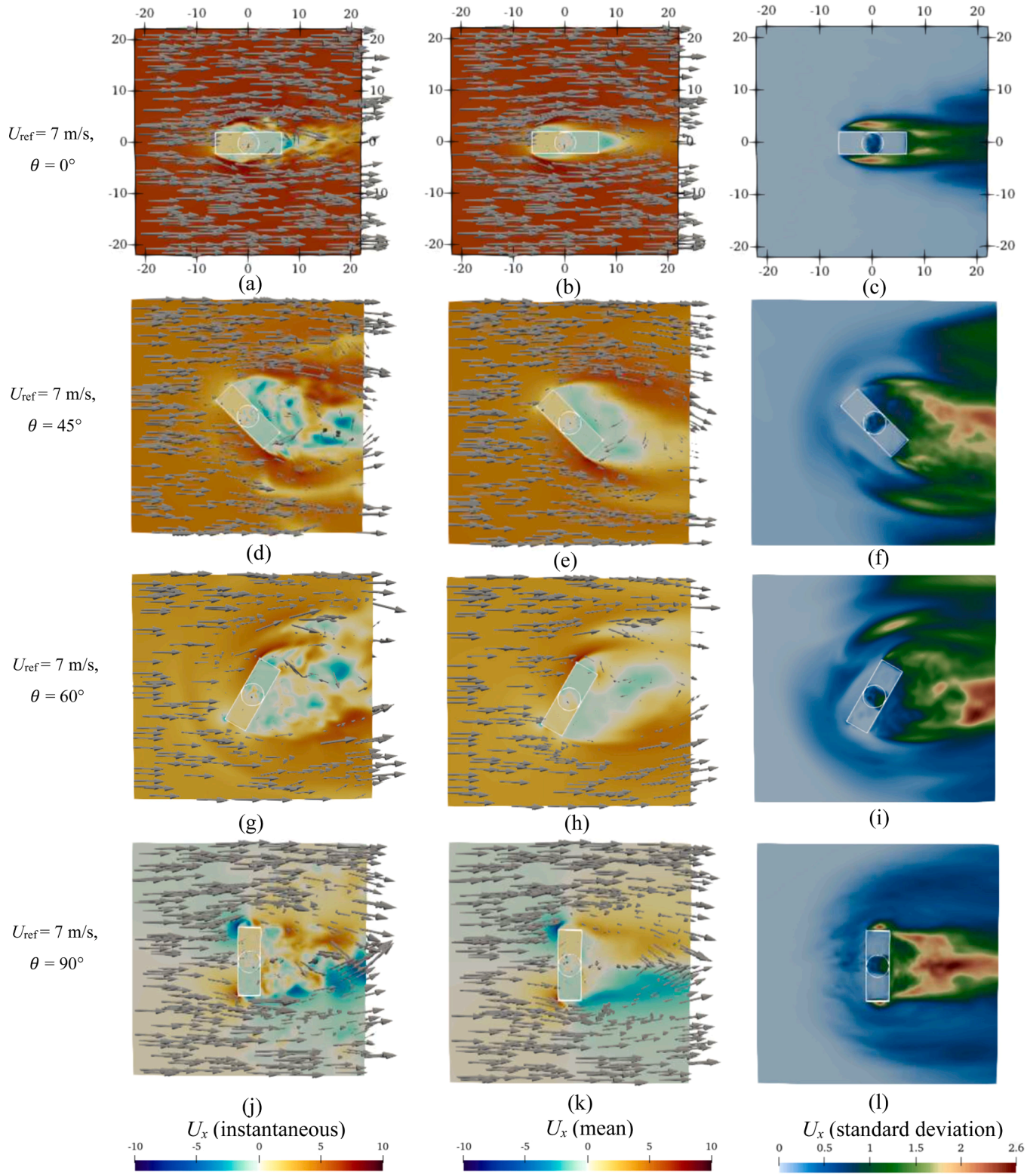


Fig. 8. Distribution of flow velocity coloured by the velocity component aligned with DAC lengthwise direction (U_x) (m/s) overlaid by vectors of flow velocity magnitude including instantaneous (at t = 155 s) (left column) mean (middle column) and standard deviation (right column) for different wind angles at a horizontal plane passing through the DAC unit middle height (z = 3.175 m) for the wind reference velocity of U_{ref} =7 m/s.

directed toward the unit inlet, the wake region is predominantly exposed to one of the unit inlets (leeward inlet), causing re-entry of a considerable amount of the CO_2-depleted plume into the DAC unit through the leeward inlet, as observed in the second row of Fig. 5 and first row of Fig. 6. By increasing the wind angle to $\theta = 45°$, the wake region is formed near the DAC leeward inlet and close to one of the DAC vertical walls. The leeward inlet is only exposed to some portion of the wake region at $\theta = 45°$(Fig. 6, second row). This reduces the amount of the CO_2-depleted plume re-entering the DAC unit through the DAC leeward inlet compared to the angle $\theta = 0°$, as observed in Fig. 5 (second

and third row). As the wind angle increases further to $\theta = 60°$, the portion of the wake region exposed to the DAC inlets is further reduced, causing a further reduction in the amount of CO_2-depleted plume re-entering the DAC unit through the DAC leeward inlet, as observed in Figs. 5 (fourth row) and 6 (third row). With a further increase in wind angle to $\theta = 90°$, the wind direction is perpendicular to the DAC unit inlet, causing the formation of almost the entire wake region at the back of the DAC vertical walls, minimising exposure of the unit inlets to the wake region (Fig. 6, fourth row). As such, almost no CO_2-depleted air plume is returned to the unit through the unit inlets at the wind angle of $\theta = 90°$ (Fig. 5, fifth row), confirming the trend observed in Fig. 4. According to Eqs. (7)–9, the CO_2 concentration at the unit inlet can affect the CO_2 capture characteristics. Therefore, since the wind direction affects the CO_2 concentration at the DAC unit inlet (Figs. 5 and 6), it is also a factor affecting CO_2 capture characteristics.

It is also worth mentioning that there is only a small variation of CO_2 concentration at the DAC inlet (less than the permissible level of 10 %), indicating the validity of Eq. (13) for the CO_2 capture rate dependency on the DAC inlet velocity in different scenarios. In our previous study (Eftekharian et al., 2025b), it was shown that, for wind velocities ranging between 0 and 9 m/s at a wind angle of $\theta = 0°$, the highest deviation in inlet CO_2 concentration occurred at a wind reference velocity of 7 m/s, with a deviation of approximately 7.5 %, which remained within the permissible range. In the current study, DAC inlet CO_2 concentration was further evaluated for different wind angles and speeds. The results indicate that the highest deviation corresponds to the same wind angle of $\theta = 0°$. This is mainly because the strongest interaction between the CO_2-depleted exit plume, wake region, and the DAC inlet occurs at $\theta = 0°$ wind angle (Figs. 5 and 6). As discussed earlier in this section, at this angle, the wake is predominantly aligned with the leeward inlet, leading to the highest exposure of the DAC leeward inlet to the CO_2-depleted plume mixed into the wake region, which results in a reduction of CO_2 concentration at the DAC leeward inlet. As the wind angle increases, the direct exposure of the leeward inlet to the wake region decreases, leading to less reduction in the DAC leeward inlet CO_2 concentration compared to the case of $\theta = 0°$. The inlet exposure to the wake region becomes minimal when the wind is perpendicular to the DAC inlet (case of $\theta = 90°$), leading to almost no variation in the DAC inlet CO_2 concentration (Figs. 5 and 6). Accordingly, the highest deviation in the DAC inlet CO_2 concentration was observed for the $\theta = 0°$ wind angle. As such, all the investigated scenarios in the current study remain within the permissible 10 % variation threshold, confirming the applicability of Eq. (13) for all considered cases. A separate analysis is required to evaluate the conditions beyond the studied wind velocity range to assess whether the associated DAC inlet CO_2 concentration variations remain within the acceptable limits, which can be the subject of future studies.

In addition to CO_2 concentration, Eq. (9) demonstrates the dependence of CO_2 capture characteristics on the unit inlet velocity. Figs. 7 and 8 show the distribution of flow velocity component aligned with DAC lengthwise direction (U_x). Fig. 7 shows that the presence and direction of wind influence the velocity distribution near the DAC unit inlets, which consequently affects the DAC CO_2 capture characteristics. For the no-wind condition, the CO_2-depleted plume rises vertically (first row in Fig. 5), and there is almost no variation in flow velocity (Fig. 7c) and CO_2 concentration (Fig. 5c) at the unit inlets. This results in constant CO_2 capture characteristics in still winds, which is reflected in the almost zero value of CO_2 standard deviation at the immediate exit of the DAC unit (Fig. 5c). However, with the introduction of wind aligned with the unit inlet ($\theta = 0°$), due to the wake recirculation zone formed near the DAC leeward inlet, temporal variations in velocity can be observed adjacent to the DAC leeward inlet (Fig. 7f and Fig. 8c). The combined effects of variations in unit inlet velocity and the CO_2 concentration at the unit inlet affects the CO_2 capture characteristics, which can be represented in the variation of CO_2 concentration at the immediate exit of the unit outlet (Fig. 5f). With the increase of wind angle to $\theta = 45°$, a relatively higher time variation of flow adjacent to the unit inlets can be observed (Fig. 7i and Fig. 8f), which leads to a slightly higher variation of CO_2 capture characteristics at the immediate exit of the unit outlet (Fig. 5i). With continued increase of wind angle to $\theta = 60°$, the time variation of flow velocity adjacent to the unit inlets is intensified (Fig. 7l and Fig. 8i) due to the stronger streamwise and horseshoe vortices formed at wind angle of $\theta = 60°$ compared to that at $\theta = 45°$, leading to a slight reduction in DAC outlet CO_2 concentration (Fig. 5, fourth row). With a further increase in wind angle to $\theta = 90$, a considerably higher variation of velocity can be observed close to the unit inlets (Fig. 7o and Fig. 8l). This is due to the flow vortex structure associated with wind striking the sharp edges of DAC unit inlets, as well as horseshoe vortices affecting flow velocity distribution at the unit inlets, which will be discussed in more detail later in this section. This high variation of flow velocity at the unit inlet directly impacts CO_2 capture characteristics, resulting in a significantly higher variation of CO_2 concentrations at the immediate exit of the DAC unit at higher wind angles, as observed in Fig. 5.

4.3. *Quantitative assessment of CO_2 capture*

We now further quantitatively examine the impacts of wind direction on the CO_2 capture characteristics of the DAC unit. The time history of the mean surface-averaged capture amount, flow velocity, and capture rate at both DAC unit inlets is plotted in Figs. 9–12. It should be noted that the unit inlet velocity (U_{in}) variable, shown in Figs. 9–12, corresponds to the surface-area-averaged velocity normal to each DAC inlet surface. For each DAC inlet, U_{in} is calculated by averaging the local normal velocity component over the inlet area. Figs. 9 and 10 present DAC inflow and CO_2 capture characteristics under the reference wind speeds of 7 m/s and 2 m/s. These figures show that when the ambient flow is still, the time history of the inlet velocity is the same at both inlets and has negligible fluctuations. Since the CO_2 capture amount and capture rate are a function of inlet velocity (Eqs. (9), 13), these values are also similar at both inlets.

With the introduction of wind, the flow velocity at both inlets as well as the CO_2 capture amount and capture rate start to fluctuate. The presence of wind also causes different fluctuation patterns and magnitudes at the inlets. Comparison of Figs. 9 and 10 demonstrates that for a given wind angle, with the increase of wind speed, the fluctuations in inlet velocity and CO_2 capture amount increase. At $\theta = 0°$, the inlet velocity and capture amount are higher at the windward inlet. This is because the leeward inlet is exposed to the wake recirculation region formed downstream of the DAC unit, causing a lower flow entrainment into the DAC unit through the leeward inlet rather than the windward inlet. According to Eqs. (8)–9, since the CO_2 capture amount is a function of DAC inlet velocity, the windward inlet contributes to a higher capture amount than the leeward inlet at the wind angle of $\theta = 0$, while the angle $\theta = 90°$ shows almost equal mean inlet velocity, capture rate, and capture amount for both inlets. This is mainly because at $\theta = 90°$, both DAC inlets are exposed to a similar vortex structure due to symmetry, as shown in Figs. 4–8. In general, the behaviour of windward and leeward inflow velocity, CO_2 capture rate, and capture amount depends on the dominant wake and vortical structures formed around each DAC inlet (Fig. 4). At the wind angle of $\theta = 0°$, the DAC leeward inlet is predominantly aligned with the wake recirculation region (Fig. 4), whereas the windward inlet is primarily subjected to the incoming flow with less vortical effect, leading to a higher contribution of the windward inlet to the DAC inflow and CO_2 capture amount (Figs. 9–10). With a slight increase in wind angle to $\theta = 30°$, the wake region remains mainly aligned with the DAC leeward inlet (Fig. 4). As such, the case with $\theta = 30°$ preserves a similar trend of DAC inlet contributions, as reflected in the DAC inlet flow velocity and CO_2 capture amount (Figs. 9 and 10). As the wind angle further increases to $\theta = 45°$ and $\theta = 60°$, particularly at higher oblique angles ($\theta = 60°$), the wake recirculation region is moved from the leeward inlet zone and tends to develop closer

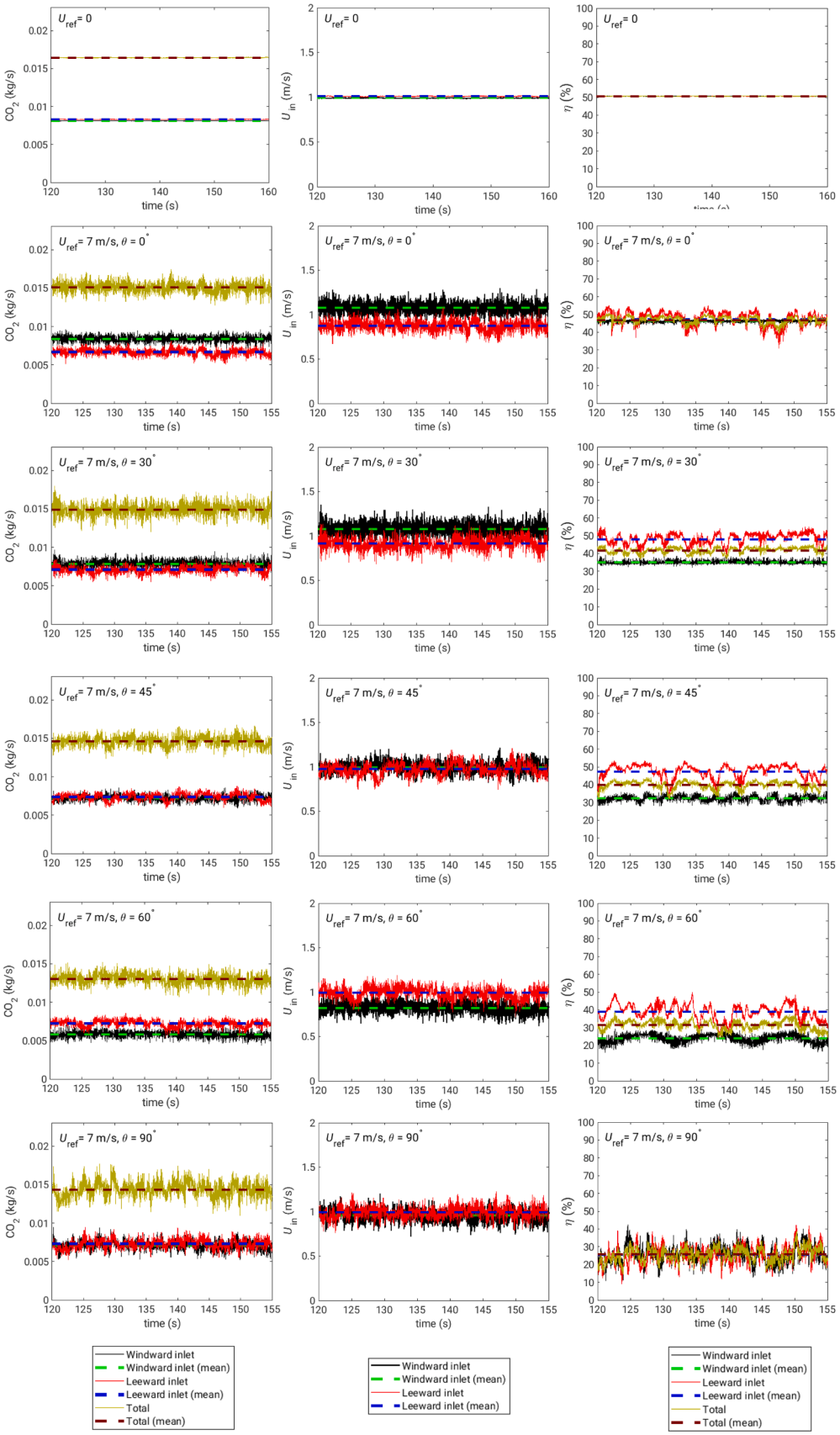


Fig. 9. Time history of surface-averaged CO_2 capture amount (kg/s) (left column), inlet velocity (U_{in}) (m/s) (middle column), and CO_2 capture rate (η %) (right column) for different wind reference angles ($\theta = 0°, 30°, 45°, 60°, 90°$) at wind speed of $U_{ref} = 7$m/s. The mean values for inlet velocity, CO_2 capture rate and capture amount are also plotted in dash lines. For all scenarios, the fan flow rate is fixed with value of 43.55 m^3/s. For $\theta = 90°$, the leeward and windward inlets in the legend indicate the right and left inlet, respectively.

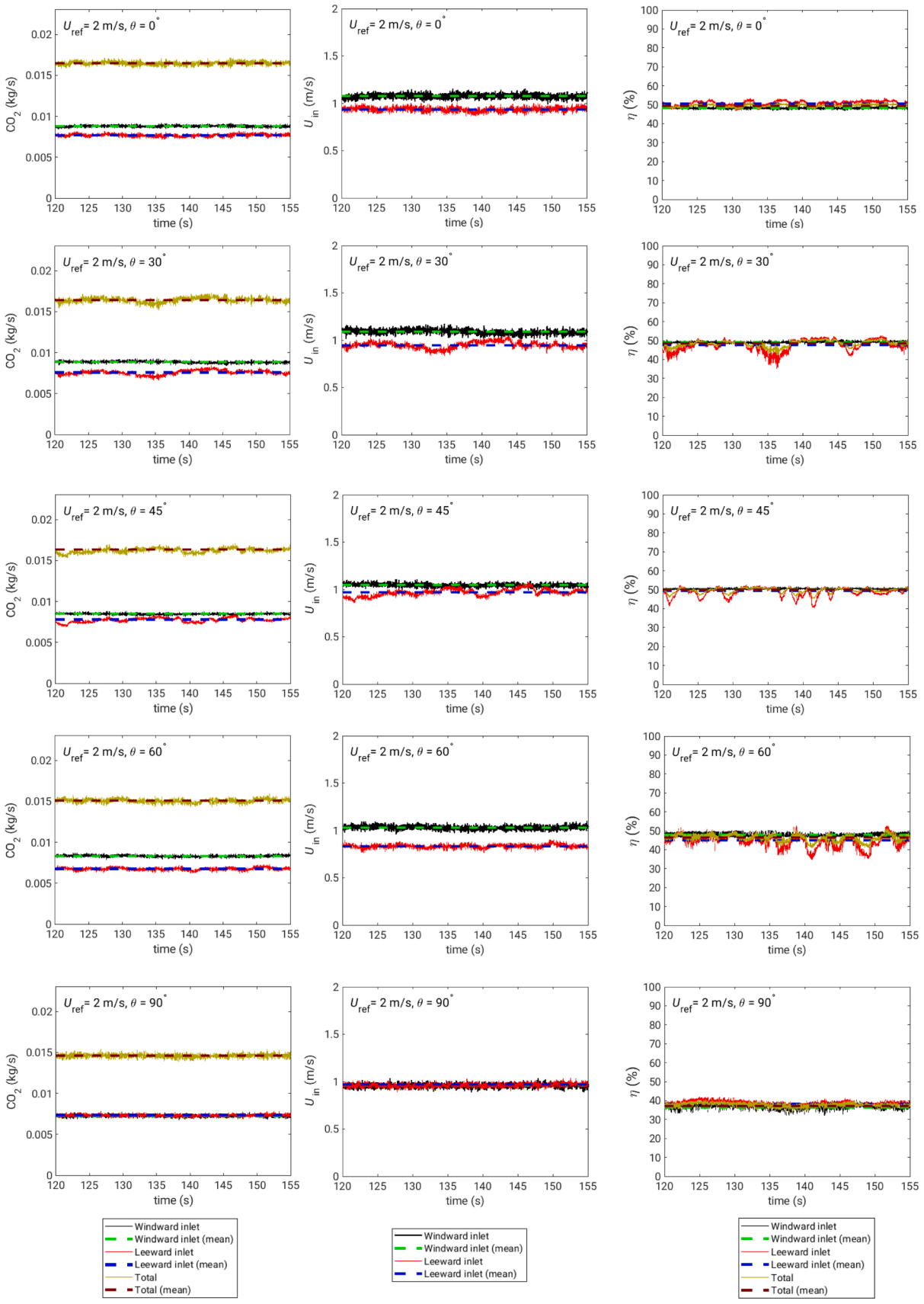


Fig. 10. Time history of surface-averaged CO_2 capture amount (kg/s) (left column), inlet velocity (U_{in}) (m/s) (middle column), and CO_2 capture rate (η %) (right column) for different wind reference angles ($\theta = 0°$, 30°, 45°, 60°, 90°) at wind speed of $U_{ref} = 2$ m/s. The mean values for inlet velocity, CO_2 capture rate and capture amount are also plotted in dash lines. For all scenarios, the fan flow rate is fixed with the value 43.55 m^3/s. For $\theta = 90°$, the leeward and windward inlet in the legend indicate the right and left inlet, respectively.

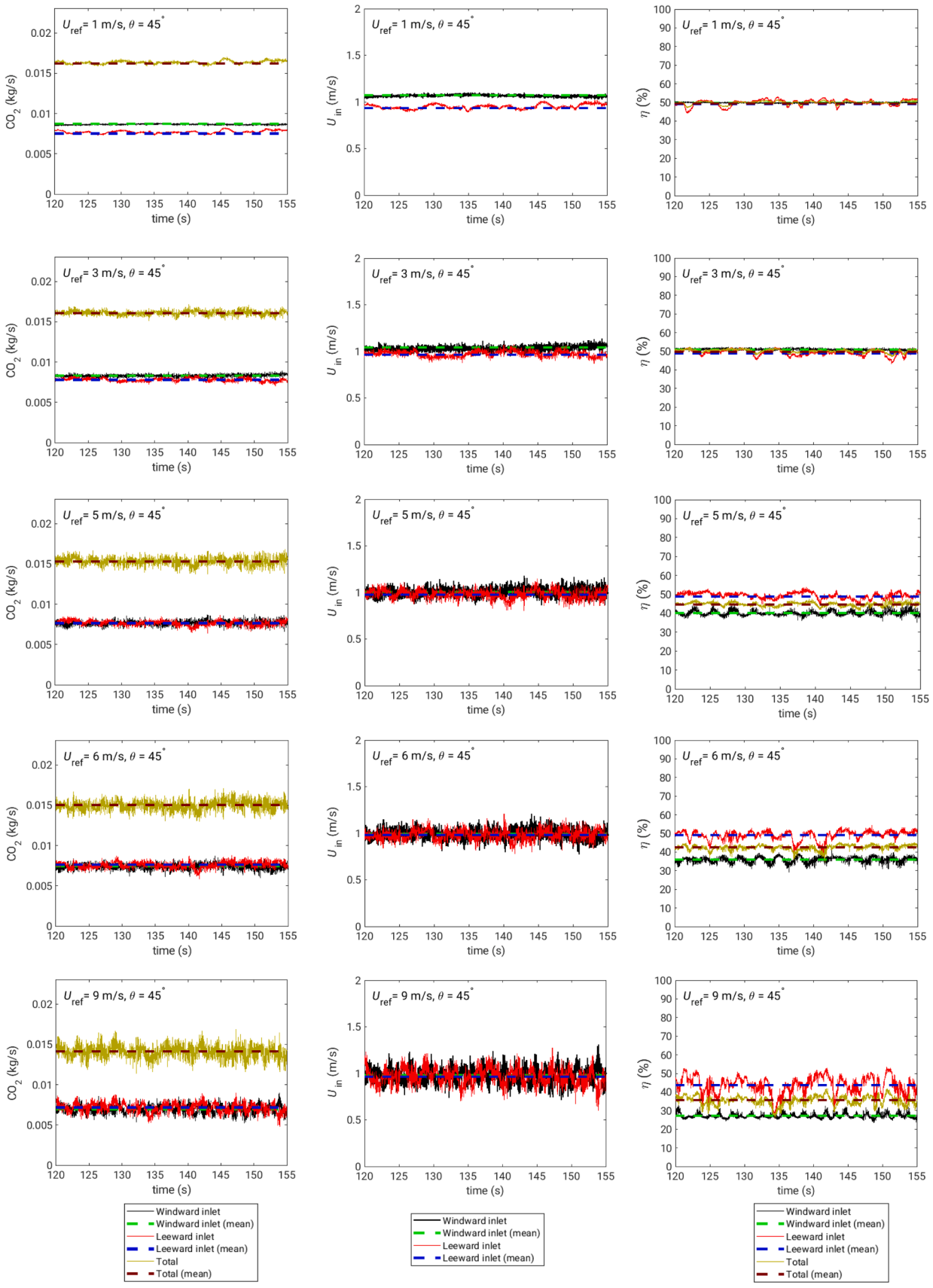


Fig. 11. Time history of surface-averaged CO_2 capture amount (kg/s) (left column), inlet velocity (U_{in}) (m/s) (middle column), and CO_2 capture rate (η %) (right column), for different reference wind speeds (U_{ref}=1, 3, 5, 6, 9 m/s) at wind angle of $\theta = 45^{\circ}$. The mean values for inlet velocity, CO_2 capture rate and capture amount are also plotted in dash lines. For all scenarios, the fan flow rate is fixed with value 43.55 m^3/s.

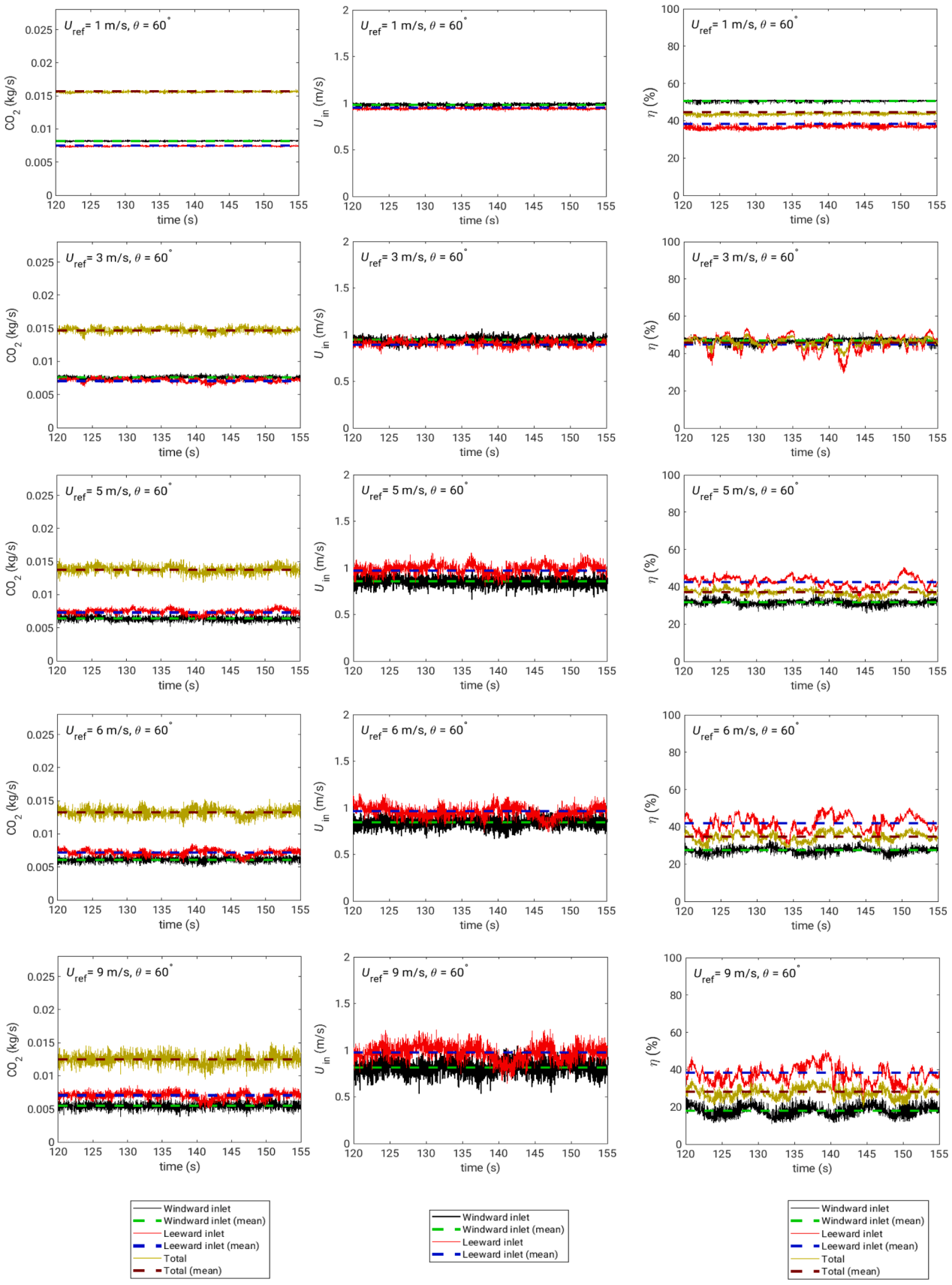


Fig. 12. Time history of surface-averaged CO_2 capture amount (kg/s) (left column), inlet velocity (U_{in}) (m/s) (middle column), and CO_2 capture rate (η %) (right column), for different reference wind speeds (U_{ref} =1, 3, 5, 6, 9 m/s) at wind angle of $\theta = 60°$. The mean values for inlet velocity, CO_2 capture rate and capture amount are also plotted in dash lines. For all scenarios, the fan flow rate is fixed with value 43.55 m^3/s.

to the DAC side wall (Fig. 4, Fig. 6 and Fig. 8). Moreover, at higher oblique angles, both inlets are strongly impacted by local shear-layer separation and flow turning effects. At $\theta = 90^{\circ}$, the wake recirculation zone is almost completely displaced toward the DAC wall (Fig. 4, Fig. 6 and Fig. 8), and due to the symmetry, both inlets are primarily affected by almost similar local shear-layer separation mechanisms. The formation of these strong shear-layer separation structures at angles of $\theta = 60^{\circ}$ and $\theta = 90^{\circ}$ significantly contributes to relatively strong CO_2 capture fluctuations at these angles at high wind speeds, as observed in Fig. 9 and Fig. 12.

Generally, increasing the wind angle from $\theta = 0^{\circ}$ towards a near-perpendicular angle results in a reduction in total capture rate, with a more pronounced reduction at higher wind speeds (e.g. $U_{ref} = 7$ m/s and $U_{ref} = 9$ m/s). The minimum values of capture rate are generally observed at $\theta = 60^{\circ}$ and $\theta = 90^{\circ}$, as shown in Fig. 9–12. A similar trend is also observed for capture amount, reaching its lowest value at either $\theta = 60^{\circ}$ or $\theta = 90^{\circ}$, depending on the wind speed. These results indicate that as the wind angle increases towards a near-perpendicular orientation, DAC unit performance generally deteriorates; however, among the studied angles, the most unfavourable condition does not necessarily correspond to $\theta = 90^{\circ}$ at all wind speeds.

At higher wind speeds, the lowest CO_2 capture amount does not necessarily occur at $\theta = 90^{\circ}$ and a lower value is observed at $\theta = 60^{\circ}$. This behaviour can be attributed to the combined effects of total CO_2 capture rate and the DAC inlet CO_2 concentration. Although total CO_2 capture rate shows similarly low values at $\theta = 60^{\circ}$ and $\theta = 90^{\circ}$ (Fig. 9), DAC inlet CO_2 concentration is lower at $\theta = 60^{\circ}$ because part of the CO_2-depleted DAC exit plume is drawn back to the DAC inlet (Figs. 5–8) due to DAC inlet-wake interaction. In contrast, at $\theta = 90^{\circ}$, the wake recirculation regions are primarily formed behind the DAC unit wall, resulting in limited exposure of the inlets to the recirculating plume and consequently a relatively higher DAC inlet CO_2 concentration is observed at $\theta = 90^{\circ}$. Therefore, under high wind speed conditions, at $\theta = 60^{\circ}$, the additional reduction in DAC inlet CO_2 concentration can outweigh the small difference in DAC CO_2 capture rate and result in a higher CO_2 capture amount at $\theta = 60^{\circ}$ than that at $\theta = 90^{\circ}$. This demonstrates that the relationship between the DAC performance and wind angle is not necessarily linear and that the most unfavourable condition can occur at a near-perpendicular angle rather than exactly at $\theta = 90^{\circ}$.

Figs. 11 and 12 highlight that at oblique angles of $\theta = 45^{\circ}$ and $\theta = 60^{\circ}$, with the increase in wind speed, the fluctuations of CO_2 capture amount and DAC inlet velocity are amplified. This is mainly because, with the increase in wind speed, turbulent fluctuations as well as vortical structures formed around the DAC inlets are magnified, leading to an increase in DAC inflow fluctuations. Since the CO_2 capture characteristics are dependent on the DAC inflow velocity (Eqs. (9) and 13), CO_2 capture characteristics also undergo higher fluctuations as ambient wind speed increases. This trend is consistent with the results of Eftekharian et al. (2025b), where the behaviour of CO_2 capture characteristics and DAC inflow velocity with wind speed at a wind angle of $\theta = 0^{\circ}$ was studied, and it was shown that the fluctuations in these parameters are amplified with the increase of ambient wind speed. Figs. 11 and 12 also demonstrate that with the increase of wind speed at wind angles of $\theta = 45^{\circ}$ and $\theta = 60^{\circ}$, the mean overall CO_2 capture amount is reduced. This behaviour is also consistent with the observation of Eftekharian et al. (2025b) where it was shown that the mean overall CO_2 capture amount reduces with the increase of wind speed when the wind direction is aligned with the DAC lengthwise direction ($\theta = 0^{\circ}$).

The fluctuating behaviour of the inlet velocity observed in the wind speed scenarios of Figs. 9–12 stems from the formation of vortical structures near the DAC unit's inlets, as demonstrated in Fig. 4. The behaviours of these vortical structures vary depending on wind speed and direction. As such, the inlet velocity and CO_2 capture characteristics behave differently under varying wind speed and direction conditions.

At the $\theta = 0^{\circ}$ angle, the dominant vortex mechanism near the windward inlet is near-ground horseshoe vortices formed upstream of the DAC unit. The vortices wrap around the sides due to the interaction of ambient wind with the unit's near-ground wall frame, as shown in Fig. 4. This can affect the distribution of flow at the lower portion of the windward inlet area. The leeward inlet, however, is exposed to more vortical structure mechanisms. The flow distribution at the leeward inlet is significantly affected by the wake vortex structure as well as the interaction of wind with the vertical jet of the CO_2-depleted exit plume. These wake-induced vortex structures significantly affect the flow field at the leeward inlet (observed in Figs. 7 and 8), leading to higher flow entrainment and CO_2 capture at the windward inlet than at the leeward inlet, as shown in Figs. 9 and 10.

In addition to horseshoe vortices, by increasing the wind angle to 30 and 45 degrees, streamwise vortices associated with flow separation at the DAC unit upstream leading-edge corner affect the velocity distribution at the windward inlet (Fig. 4). At these wind angles, the windward inlet is exposed to more vortical structure mechanisms compared to the 0° angle. Moreover, the flow distribution at the leeward inlet is primarily affected by oblique wake vortices formed downstream of the inlet due to the interaction of wind with the unit as well as CO_2-depleted exit plumes. These differences in the vortical structure mechanisms near the DAC inlet between these wind oblique angles and the 0° angle result in a change in the flow field and the subsequent carbon capture characteristics associated with the unit inlets, as observed in Figs. 9 and 10.

By increasing the wind angle to 60° and 90°, streamwise edge vortices are the dominant vortex structure mechanism affecting the DAC inlets. These structures form because of the formation of flow separation when the wind strikes the vertical edges at the windward wall face, as observed in Fig. 4. Moreover, horseshoe vortices formed at the base of the DAC unit can be deflected sideways, affecting the flow field at both inlets. The formation of a wake recirculation zone downstream of the DAC unit can also have an impact on the leeward inlet at the 60-degree angle. This wake region can also affect the flow field distribution of both inlets at the 90-degree wind angle through suction effects of wake-induced vortices formed in the wake region. These strong vortex structures near the DAC inlets at high wind angles of 60° and 90° cause significant velocity fluctuations at the inlets, substantially affecting the corresponding CO_2 capture amount as observed in Figs. 7–12.

Although different wind angles might bring about some common mechanisms of vortical structure, the dominant mechanism and vortex strength are different at each angle for each inlet. This results in different flow and carbon capture characteristics at each inlet for different wind angles, as observed in Figs. 9 and 10.

5. Discussion and concluding remarks

To understand the combined effects of ambient wind speed and wind direction on the DAC performance associated with CO_2 capture efficiency, we calculate the CO_2 capture amount over a 1-year period. After considering the annual maintenance time, we estimate that the DAC unit could be on operation for approximately 8000 h per year. The total CO_2 capture amount was calculated using Eq. (10) for different combinations of wind speeds and wind angles, as shown in Fig. 13. This figure shows that increasing wind speeds reduces the CO_2 capture amount at all wind angles. This can be attributed to the increasing velocity fluctuations at the DAC inlets with increasing wind speeds for a given wind angle, as shown in Figs. 9–12. This increase in inlet flow velocity fluctuation with wind speed negatively impacts the CO_2 capture amount, as observed in Figs. 11 and 12.

Wind direction significantly impacts annual CO_2 removal (Fig. 13). For wind angles up to 45 degrees, the results for all wind speeds follow a similar trend, where increasing wind angles reduce the annual CO_2 removal. At a 60-degree wind angle, the CO_2 removal at all wind speeds experiences a relatively sharper reduction. For lower wind speeds (e.g. $U_{ref} = 3$ m/s), the annual CO_2 capture amount continues to reduce, reaching a minimum at 90 degrees, whereas at higher wind speeds (e.g. $U_{ref} = 7$ m/s and $U_{ref} = 9$ m/s), the minimum CO_2 capture amount occurs

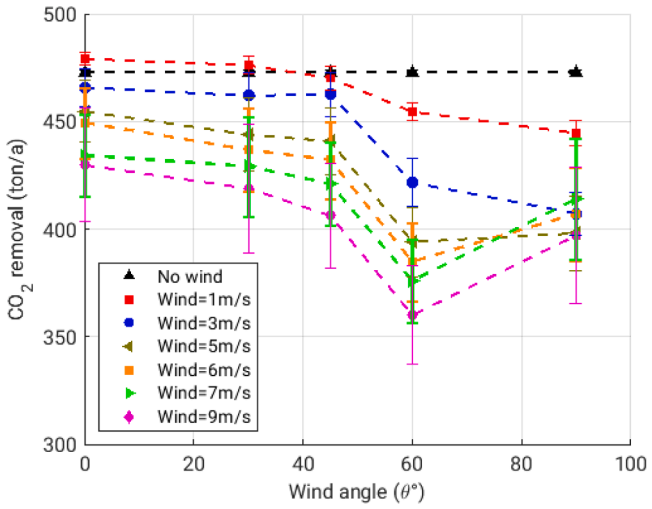


Fig. 13. CO_2 removal over 1 year (ton/a) for different wind speeds and directions. Dash lines indicate the trend, and the error bar indicates standard deviation.

somewhere between 45 and 90 degrees (around 60 degrees). Fig. 13 also shows that with increasing wind speeds and directions, the standard deviation of the annual CO_2 removal increases. Overall, for all wind speeds, the minimum annual CO_2 capture amount corresponds to near-perpendicular angles. This reduction in CO_2 removal can be attributed to the strong vortical structure formed adjacent to the DAC inlets due to the interaction of wind with the DAC unit's sharp edges, and the wake region formed downstream of the unit as shown in Fig. 4. These vortical structures affect the flow velocity distribution at the inlet, which results in a reduction of the CO_2 capture amount for higher wind angles.

In summary, the impacts of wind speed and direction on the performance of a DAC unit associated with the evaluation of instantaneous and mean CO_2 capture characteristics were studied using LES. Combinations of different atmospheric boundary layer wind speeds and directions were studied, and the corresponding CO_2 capture characteristics were extracted using our atmosDacBoundary solver for CO_2 concentration at the DAC exit. We showed that the impacts of wind speed and direction on DAC performance are significant. Generally, an increase in wind speed degrades the CO_2 capture performance. Moreover, with increasing wind angles with respect to the unit inlet, the DAC CO_2 capture amount generally reduces, with a minimum occurring when the ambient wind angle is near-perpendicular to the DAC unit inlet (e.g. wind angles of 60 and 90 degrees). This study assists with the prediction of DAC performance in different climatic conditions. Moreover, considering the related wind meteorological data, the findings of this study can inform the design orientation of a DAC unit that will yield the highest CO_2 capture performance in any potential locations.

It should be noted that the simulations conducted in the current study consider five different wind angles and therefore do not determine the exact DAC angular position corresponding to the minimum CO_2 capture amount. Instead, the results of this study demonstrate how DAC performance varies with the investigated wind directions and demonstrate the physics and mechanisms underlying the observed angle-dependent behaviour. To determine the precise wind angle corresponding to the minimum CO_2 capture amount, a finer angular resolution, particularly between $\theta = 60^\circ$ and $\theta = 90^\circ$, would be required, which can be the subject of future studies. It should also be highlighted that the quantitative performance variation analysis presented in the current study is dependent on the specific DAC geometry as well as environmental and operating conditions considered herein. Yet, the flow mechanisms and general trends associated with crosswind effects demonstrated in this study are expected to remain relevant for similar crossflow DAC configurations. Moreover, the present work concentrates on DAC airflow and CO_2 capture performance characteristics, while detailed fan energy consumption analyses are recommended for future studies.

Furthermore, additional experimental studies are required to determine the impacts of temperature, turbulent intensity, and humidity on the CO_2 capture characteristics, which can be the subject of future studies that may result in modification of the CO_2 capture-DAC inflow relationship reported in Section 2.2. However, the overall methodology implemented in the current study would remain relevant, given that an appropriate relationship between capture rate and DAC inlet condition is developed for the corresponding operating conditions.

CRediT authorship contribution statement

Esmaeel Eftekharian: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Vassili Kitsios:** Writing – review & editing, Resources, Conceptualization. **Ali Kiani:** Writing – review & editing, Resources, Conceptualization. **Ashok K. Luhar:** Writing – review & editing, Resources, Conceptualization. **Paul Feron:** Writing – review & editing, Resources, Conceptualization. **Aaron W. Thornton:** Writing – review & editing, Resources, Conceptualization. **Kathryn M. Emmerson:** Writing – review & editing, Resources, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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